# The Effect of Artificial Intelligence Exposure on Labor Productivity

Kseniia Nikitina

NYU MA Economics

## Abstract

Does exposure to artificial intelligence raise labor productivity? I study 59 U.S. industries from 1997 to 2024, combining a time-varying measure of industry exposure to AI, built from occupational ability scores and annual employment shares, with an economy-wide index of AI capability. In two-way fixed-effects regressions, neither early-stage (2016–2019) nor generative-AI-era (2020–2024) exposure has a statistically significant effect on labor productivity. The preferred estimate for the generative-AI era is 0.023 (standard error 0.364), and the confidence intervals include both sizable gains and sizable losses. Because most of the identifying variation comes from 2022–2024, the results show that any productivity effect of generative AI was not yet detectable in industry data through 2024, not that no effect exists. Significant positive estimates appear only in specifications without year fixed effects or with a lagged dependent variable, which I argue are not credible.

## 1. Introduction

The spread of artificial intelligence has placed a fundamental question of labor economics on the agenda: does new technology complement workers or displace them (Autor 2015), and what does this mean for labor productivity? Earlier waves of computerization mainly substituted for routine tasks (Autor, Levy, and Murnane 2003), whereas large language models (LLMs), the core of today's commercial AI products, reach tasks across the wage distribution, and exposure to them rises with wages (Eloundou et al. 2023; Felten, Raj, and Seamans 2023). The magnitude of the resulting productivity effect matters for aggregate growth and employment policy, but theory does not pin it down. In the task-based framework of Acemoglu and Autor (2011), new technology can either complement or displace labor, and Acemoglu and Restrepo (2019) show that automation can reduce labor demand even as it raises productivity unless new tasks reinstate labor. The question is therefore empirical.

This paper presents new evidence for the United States using a panel of 59 industries from 1997 to 2024. I combine cross-industry differences in structural AI exposure with the rapid, economy-wide improvement in AI capability to test whether more exposed industries saw faster productivity growth as AI improved. The analysis distinguishes two stages of AI development. The early stage (2016–2019) is measured with the aggregate AI Industry Exposure (AIIE) index, and the mature stage (2020–2024) with an AIIE index specific to large language models, which captures the rise of generative AI.

I find no statistically significant effect of AI exposure on labor productivity in either stage. In the preferred specification, the mature-stage coefficient is 0.023 (standard error 0.364), and its confidence interval includes both sizable gains and sizable losses. Because most of the

identifying variation comes from 2022–2024, the results show that any productivity effect of generative AI was not yet detectable in industry data through 2024, not that no effect exists.

## 2. Literature Review

### 2.1 Theoretical Foundations

The debate over whether technology complements or replaces workers goes back at least to Hicks (1932), who classified inventions as labor-saving, capital-saving, or neutral. Acemoglu and Autor (2011) gave the question its most tractable modern form through a task-based framework: rather than asking whether technology is broadly good or bad for labor, they ask which tasks machines can perform. Their model explains job polarization, in which the automation of routine middle-skill work hollows out the middle of the wage distribution while employment grows at both the top and the bottom. Acemoglu and Restrepo (2018, 2019) extend this framework by distinguishing a displacement effect, in which machines take over existing tasks, from a reinstatement effect, in which new labor-intensive tasks emerge. The key implication is that rapid automation without enough new task creation shifts the gains from higher productivity toward capital rather than workers. This framework motivates measuring AI exposure through the tasks and abilities that AI can perform, which is how the measures below are built.

### 2.2 Measuring AI Exposure

A widely used exposure measure is the AI Occupational Exposure (AIOE) index of Felten, Raj, and Seamans (2019, 2021), which links ten AI applications to the 52 occupational abilities in O*NET. Because it is built from measures of AI capabilities rather than from adoption, it is not driven by the outcomes of any particular occupation or industry. Webb (2020) and Brynjolfsson,

Mitchell, and Rock (2018) build alternative exposure measures from the overlap between patent text and job-task descriptions and from crowdsourced ratings of work activities, respectively. This study extends the Felten et al. (2021) index to a measure of AI exposure that varies across both industries and years.

### 2.3 Empirical Evidence on AI and Productivity

The empirical record is mixed. Plikas, Trakadas, and Kenourgios (2024) find a significant positive relationship between AI technologies and GDP across EU27 countries over 2010–2020, and that employing information and communication technology (ICT) professionals and reskilling employees are associated with higher labor productivity. For an earlier automation technology, Graetz and Michaels (2018) find that robots raised annual labor productivity growth in 17 countries by about 0.36 percentage points over 1993–2007. Brynjolfsson, Rock, and Syverson (2019) argue that the gains from AI may take longer to appear, because transformative technologies require costly complementary investments and reorganization first. This implementation-lag argument motivates my distinction between pre-AI, early, and mature periods.

On the labor side, automation brings displacement as well as gains. Acemoglu and Restrepo (2020) find that one more robot per thousand workers reduced the employment-to-population ratio by 0.2 percentage points and wages by 0.42 percent across U.S. commuting zones, while Graetz and Michaels (2018) find no significant effect on total employment. At the firm level, Acemoglu, Lelarge, and Restrepo (2020) find that French manufacturing firms that adopt robots expand value added and employment at the expense of their competitors, so adoption is

negatively associated with employment overall. Seamans and Raj (2018) argue that the use of AI and robotics needs to be measured systematically at the firm level.

For the LLM era, Eloundou et al. (2023) estimate that LLMs could let workers complete about 15 percent of U.S. work tasks significantly faster at the same quality. Experimental and field evidence finds large gains on individual tasks: ChatGPT cut the time professionals needed for writing tasks by 40 percent and raised output quality by 18 percent (Noy and Zhang 2023), and an AI assistant raised customer-support agents' productivity by 15 percent on average, most for less experienced and lower-skilled workers (Brynjolfsson, Li, and Raymond 2025). Economy-wide effects have been small so far: two years after the launch of ChatGPT, Humlum and Vestergaard (2025) rule out effects larger than 2 percent on earnings or recorded hours in Denmark.

### 2.4 Positioning This Study

This paper sits at the intersection of research on AI and aggregate productivity (Brynjolfsson, Rock, and Syverson 2019; Plikas et al. 2024), work on occupational exposure to AI (Webb 2020; Felten, Raj, and Seamans 2021), and the emerging LLM literature (Eloundou et al. 2023; Humlum and Vestergaard 2025). Existing exposure measures are mostly cross-sectional, while macro studies typically rely on a single aggregate time series. This paper combines the two: it builds an industry-level exposure measure that varies over time and scales it by an economy-wide index of AI capability, which allows separate estimates for the early and mature stages of AI in a two-way fixed-effects panel.

## 3. Methodology

This paper aims to build a two-way fixed effects model to determine whether there is any empirical evidence that the introduction of generative AI across industries in the USA has had an impact on labor productivity. Conceptually, my model builds on classical foundations of labor economics and productivity theory but also links the recent developments of the task-based view of automation developed by Acemoglu and Restrepo (2018, 2019, 2022) with the continuously evolving goal of measuring the impact of generative AI on the economy. This paper provides a complementary, reduced-form empirical counterpart to these studies, since I use them to define the way AI is captured as a variable for each occupation via the skill and task displacement features developed by this literature. Previously, Stiroh (2002) and Jorgenson, Ho, and Stiroh (2008) documented the relationship between ICT-capital deepening and labor-productivity growth across U.S. industries using BLS productivity decompositions, which act as a skeleton for explaining productivity in the first half of my panel, from 1997 until 2016. This is when more sophisticated algorithms and broader machine learning began to be introduced into many jobs. Studies of robotics that use panel data to estimate the effects of robot adoption on productivity, employment, and wages, such as Graetz and Michaels (2018) for industries in 17 countries and Acemoglu and Restrepo (2020) for U.S. local labor markets, are where I borrow controls and panel methodology from. Finally, for the last piece of the model, I adapt an existing AI exposure metric, developed from the same task framework by Felten et al. (2021), to allow for dynamic AI exposure scoring in the workplace.

### 3.1 Data

My panel covers 60 U.S. industries categorized by the NAICS system over the 28 years from 1997 to 2024. Occupation- and task-level data are merged using 4-digit NAICS or BEA codes and then aggregated to the industry level. The data come from several sources, including the Bureau of Labor Statistics (BLS) and O*NET, and amount to over a million observations at the most granular level. I exclude Farms because of missing observations, and a few other industry-year observations are also missing, so the regressions use 59 industries and 1,649 industry-year observations.

### 3.2 Dependent Variable

The dependent variable is the log of the BLS labor productivity index (2017 = 100), ln(LP), which measures output per hour worked in each industry. Appendix VII shows how labor productivity evolves within industries over the sample.

### 3.3 Measuring AI Exposure

The most consequential modeling choice is how AI exposure is measured. I build on the AIOE of Felten, Raj, and Seamans (2019, 2021) and extend it in three ways: I let it vary over time, I construct a separate index for language modeling, and I scale it by the capability of frontier AI systems. The original AIOE starts from an Amazon Mechanical Turk survey that rates how related each of ten common AI applications, such as image recognition, image generation, language modeling, speech recognition, and translation, is to each of the 52 O*NET occupational abilities. Summing these relatedness scores across applications gives an exposure score for each

$$\mathrm{AIOE_k} = \frac{\sum_{j=1}^{52}(A_{ij} \times L_{jk} \times I_{jk})}{\sum_{j=1}^{52}(L_{jk} \times I_{jk})}$$

ability. Occupation-level exposure then averages the ability scores, weighting each ability by its prevalence and importance in the occupation:

where *A* is the ability exposure score, *L* and *I* are the O*NET prevalence and importance scores, *j* indexes abilities, and *k* indexes occupations.

Felten, Raj, and Seamans (2021) aggregate occupational exposure to the industry level using employment weights from a single year. Because I need the measure to vary within industries over time, so that its effect can be estimated with industry and year fixed effects, I reconstruct industry exposure annually using BLS Occupational Employment and Wage Statistics (OEWS) national industry-specific data for 2016–2024, which report detailed occupational employment for each 4-digit NAICS industry. For each industry and year, AIIE is the average of occupational exposure weighted by each occupation's share of the industry's employment:

$$\mathrm{AIIE_{i,t}} = \sum_k \frac{\mathrm{emp_{ik,t}}}{\sum_k \mathrm{emp_{ik,t}}} \times \mathrm{AIOE_{k,t}}$$

Changes in AIIE within an industry therefore reflect changes in its occupational composition. One limitation is that I hold the O*NET ability scores fixed at their 2020 values, so these compositional changes are the only source of within-industry variation. Archived O*NET releases could be used to let the ability scores vary over time.

The release of GPT-3 in 2020 marked a shift in the technological frontier toward generative models. Following Felten, Raj, and Seamans (2023), who adapt the AIOE to language modeling, and the related LLM exposure measure of Eloundou et al. (2023), I construct two versions of industry exposure: Aggregate AIIE, which covers all ten AI applications, and LM AIIE, which uses an updated mapping that weights abilities by their relatedness to language modeling.

I divide the AI era into two periods. Early AI (2016–2019) is marked by the introduction of machine-learning language models developed by Google, and Mature AI (2020–2024) begins with the public release of GPT-3 in 2020 and includes the launch of ChatGPT in late 2022. With the years before 2016, the panel therefore has three periods: pre-AI (1997–2015), Early AI, and Mature AI. I interact the Early indicator with Aggregate AIIE and the Mature indicator with LM AIIE.

Finally, an industry with a given exposure faced far less capable AI in 2016 than in 2024. I therefore scale exposure by Intel, an index of frontier AI capability based on the Artificial Analysis Intelligence Index (Artificial Analysis 2026), which tracks the performance of frontier AI models on standardized benchmarks, supplemented by LLM token-pricing data from Demirer, Fradkin, Tadelis, and Peng (2025). Intel is normalized to lie between 0 and 1 and set to 0 before 2016, when no commercially accessible AI tools of material economic consequence existed at scale.

### 3.4 Model Specification

I estimate the following two-way fixed-effects model:

$$\ln LP_{i,t} = \beta_1 (Early \times Agg \times Intel)_{i,t} + \beta_2 (Mature \times LM \times Intel)_{i,t} + \gamma' X_{i,t} + \alpha_i + \delta_t + u_{i,t}$$

where $i$ indexes the 59 industries in the estimation sample and $t$ the years 1997–2024. The terms $\alpha$ and $\delta$ are industry and year fixed effects, and $X$ contains the ICT-capital, non-ICT-capital, and energy contributions described in Section 3.6. The preferred specification (Table 1, column 9) also includes the WFH × COVID interaction described in Section 3.7; the baseline (column 5) omits it. The coefficients on the two AI terms measure how much higher labor productivity is in more exposed industries as AI capability grows, in the early and mature periods respectively. Standard errors are clustered by industry (Arellano 1987), and the diagnostic tests in Section 4.3 support the two-way specification.

### 3.5 Identification

Because Early, Mature, and Intel vary only over time, each AI term equals industry exposure multiplied by a common time path. The year fixed effects absorb any effect of AI that is common to all industries, so the coefficients are identified only from differences across industries in how productivity changed as AI capability rose. Intel adds no cross-industry variation; it determines how much weight each year's cross-industry differences receive. The design is therefore similar to a difference-in-differences with a continuous treatment, or to a shift-share design that combines cross-sectional exposure with an aggregate shock. With a single common shock, identification rests on exposure being unrelated to other determinants of productivity growth (Goldsmith-Pinkham, Sorkin, and Swift 2020), rather than on many independent shocks (Borusyak, Hull, and Jaravel 2022). The key assumption is that, absent AI, more and less exposed industries would have followed parallel productivity paths after accounting for the fixed effects and controls. With a continuous treatment, comparing effects across exposure levels requires a stronger version of this assumption (Callaway, Goodman-Bacon, and Sant'Anna 2024).

The construction of the AI terms limits, but does not remove, simultaneity. The ability scores are fixed economy-wide and Intel is common to all industries, so neither responds to productivity in a single industry. However, the employment shares used to build AIIE are updated each year, so exposure is not strictly predetermined: an industry whose productivity is rising may also be shifting toward more exposed occupations. Because AIIE varies over time, its main effect is also not absorbed by the industry fixed effects, and the model does not include it separately. Holding the shares at their 2016 values would address both issues.

Two further limitations remain. First, I do not test for differential pre-trends. The pre-2016 years contribute to the fixed effects and controls but not to the AI coefficients, and an event-study specification that interacts exposure with individual years would show whether exposed industries were already on different productivity paths before 2016. Second, the identifying variation for the Mature term is concentrated in 2022–2024 (Appendix V), a short window that follows the COVID-19 shock. Section 3.7 describes how I control for the pandemic.

### 3.6 Control Variables

Three controls absorb important non-AI determinants of labor productivity. The capital contribution to labor productivity is split into ICT and non-ICT components, each measured as the capital service flow attributable to that asset class weighted by its share of capital income. ICT capital includes computers, software, and communication equipment, while non-ICT capital includes other equipment, machinery, and structures. Energy intensity enters as the contribution of energy inputs to labor productivity growth, measured as the value of energy services consumed per unit of output weighted by energy's share of total input costs. Because AI may

raise productivity partly through investment in ICT capital, the ICT-capital control may absorb part of any AI effect.

### 3.7 Remote Work and the COVID-19 Shock

The Mature period overlaps with the COVID-19 pandemic, which changed measured labor productivity for reasons unrelated to AI. Job losses in 2020 were concentrated among lower-productivity workers, which can raise average output per hour, and the feasibility of remote work differed widely across industries (Dingel and Neiman 2020; Bick et al. 2023). The industries where remote work is most feasible, such as legal services and securities, are also among the most exposed to language models (Appendices V and IX), so without a control the Mature coefficient could partly capture the pandemic shock. I therefore add a work-from-home (WFH) feasibility index built from the occupational classification of Dingel and Neiman (2020), interacted with a COVID indicator equal to one in 2020 and 2021. Each occupation is classified as feasible (1) or not feasible (0) to perform at home, and the index averages these values across the occupations in each industry, giving a value between 0 and 1. Because remote work persisted after 2021 (Bick et al. 2023), this control captures only the acute phase of the shock.

### 3.8 Dynamic Specifications

Labor productivity is highly persistent, so a natural extension of the static model is to add the lag of ln(LP). I also tried other lagged controls: lagged AI interactions, lagged ICT capital, and lagged employment. Some were highly collinear with other regressors or with the industry fixed effects and were excluded (Appendices III, IV and VI). I keep the static model as my preferred specification for two reasons. First, with a lagged dependent variable the AI coefficients measure one-year effects, and the implied long-run effects are implausibly large (Section 4.2). Second,

within-group estimates with a lagged dependent variable are biased (Nickell 1981), although the bias shrinks as the number of years grows. I report the dynamic specifications for transparency; a dynamic-panel GMM estimator would be a natural extension.

## 4. Results

### 4.1 Main Estimates

Table 1 builds up from pooled OLS (column 1) to the preferred specification. Column 5 is the static two-way fixed-effects model, and column 9, my preferred specification, adds the WFH × COVID control. Columns 6–8 add lagged terms to the column 9 specification and are discussed in Section 4.2.

In the preferred specification, the coefficient on Mature × LM AIIE × Intel is 0.023 with a standard error of 0.364, so its 95 percent confidence interval runs from about −0.69 to 0.74. The coefficient on Early × Aggregate AIIE × Intel is −0.801 (standard error 1.666). Neither coefficient is statistically significant, and both intervals are wide enough to include sizable gains and sizable losses, so the data cannot distinguish a meaningful productivity effect of AI from no effect. Without the WFH control (column 5), the estimates are nearly identical (0.011 and −0.746).

The WFH × COVID coefficient is negative but insignificant (−0.071, standard error 0.153), and adding it barely changes the AI coefficients, so differences in remote-work feasibility during 2020–2021 do not appear to drive the results. The three input controls are positive and significant.

The benchmark specifications in columns 1–4 also produce insignificant Mature coefficients, with one exception: in first differences (column 2), the coefficient is 0.172 (standard error 0.052). That specification has no year fixed effects, so it does not account for economy-wide shocks in the same years, and I do not treat it as causal. The tests in Section 4.3 support the two-way specification. For a descriptive view, Appendix VIII plots raw changes in employment and labor productivity by industry over 2016–2022.

# Table 1

AI Exposure and Labour Productivity — Full panel (1997–2024) build-up (clustered SEs in parentheses)

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) |
|---|---|---|---|---|---|---|---|---|---|
| | **Pooled OLS** | **First Diff** | **Industry FE** | **Year FE** | **Two-way FE** | **+ LP lag** | **+ LP lag + emp.** | **+ emp. lag** | **BASE + WFH** |
| Early × Aggregate AIIE × Intel | −0.100 | 0.084 | −1.225 | 0.017 | −0.746 | 0.025 | 0.276 | 0.192 | −0.801 |
| | (0.340) | (0.206) | (1.315) | (0.392) | (1.554) | (0.291) | (0.191) | (0.308) | (1.666) |
| Mature × LM AIIE × Intel | 0.132 | 0.172*** | −0.076 | 0.169 | 0.011 | 0.091*** | 0.134*** | 0.204 | 0.023 |
| | (0.178) | (0.052) | (0.340) | (0.204) | (0.389) | (0.028) | (0.043) | (0.148) | (0.364) |
| WFH index × COVID dummy | | | | | | −0.004 | 0.014 | 0.019 | −0.071 |
| | | | | | | (0.027) | (0.020) | (0.042) | (0.153) |
| ICT-capital contribution | 0.027*** | 0.018*** | 0.029*** | 0.024*** | 0.024*** | 0.001 | 0.006 | 0.015*** | 0.024*** |
| | (0.002) | (0.002) | (0.003) | (0.003) | (0.004) | (0.001) | (0.004) | (0.005) | (0.004) |
| Non-ICT capital contribution | 0.010** | 0.002 | 0.011*** | 0.009 | 0.009** | 0.001 | 0.000 | 0.005 | 0.009** |
| | (0.005) | (0.005) | (0.004) | (0.005) | (0.004) | (0.001) | (0.005) | (0.007) | (0.004) |
| Energy contribution | 0.012* | 0.011*** | 0.018** | 0.013** | 0.022** | 0.003* | 0.019 | 0.039** | 0.022** |
| | (0.006) | (0.004) | (0.009) | (0.007) | (0.009) | (0.002) | (0.012) | (0.019) | (0.009) |
| ln(LP)_t-1 | | | | | | 0.945*** | 0.663*** | | |
| | | | | | | (0.038) | (0.076) | | |
| Log employment_t-1 | | | | | | | −0.015 | −0.096 | |
| | | | | | | | (0.016) | (0.065) | |
| Constant | 2.389** | 0.010*** | | | | | | | |
| | (0.939) | (0.002) | | | | | | | |
| Observations | 1649 | 1590 | 1649 | 1649 | 1649 | 1590 | 528 | 528 | 1649 |
| $R^2$ | 0.410 | 0.114 | 0.447 | 0.335 | 0.344 | 0.904 | 0.709 | 0.311 | 0.344 |
| Adj. $R^2$ | 0.408 | 0.111 | 0.425 | 0.322 | 0.306 | 0.898 | 0.661 | 0.201 | 0.306 |
| Industry fixed effects | No | Differenced out | Yes | No | Yes | Yes | Yes | Yes | Yes |
| Year fixed effects | No | No | No | Yes | Yes | Yes | Yes | Yes | Yes |
| Sample years | 1997–2024 | 1997–2024 | 1997–2024 | 1997–2024 | 1997–2024 | 1998–2024 | 2017–2024 | 2017–2024 | 1997–2024 |
| SE cluster | Industry | Industry | Industry | Industry | Industry | Industry | Industry | Industry | Industry |

* p < 0.1, ** p < 0.05, *** p < 0.01

Dependent variable: ln(LP) — log of the BLS labour productivity index.

Columns (1)–(5) drop both the LP lag and lagged employment, allowing the full 1997–2024 panel.

Column (6) adds ln(LP)_{t-1}; first-year observations drop because of the lag.

Column (7) adds both ln(LP)_{t-1} and log(employment)_{t-1}; lagged employment is unavailable pre-2016, mechanically reducing the sample to 2017–2024.

Column (8) adds only log(employment)_{t-1}; same 2017–2024 restriction as (7) but without the LP lag (matches the original 2016–2024 baseline).

Pre-2016 observations contribute zero to the AI interactions by construction.

Standard errors clustered at the industry level (HC1).

Column (9) augments specification (5) with WFH × COVID, where COVID = 1 for 2020–2021 and 0 otherwise. The COVID dummy alone is absorbed by year FE; identification of the interaction comes from cross-industry variation in WFH exposure during the 2020–2021 window.

### 4.2 Dynamic Specifications

Columns 6 and 7 add the lagged dependent variable; column 7 also adds lagged employment, which shortens the sample. The Mature coefficient becomes positive and significant: 0.091 (standard error 0.028) in column 6 and 0.134 (0.043) in column 7. I do not interpret these estimates as evidence of an AI effect. With a lagged dependent variable, the coefficient measures a one-year effect, and with a persistence coefficient of 0.945 (column 6), the implied long-run effect is $0.091/(1 - 0.945) \approx 1.65$ log points, which is implausibly large. Within-group estimates with a lagged dependent variable are also biased (Nickell 1981), and lagged employment is nearly collinear with the industry fixed effects (Appendix VI). Finally, the result depends on the lag itself: column 8, which uses the same sample as column 7 but drops the lagged dependent variable, gives an insignificant 0.204 (standard error 0.148).

### 4.3 Diagnostic Tests

*Fixed effects.* An F-test strongly rejects pooled OLS in favor of industry fixed effects ($F = 9.984$, $p = 6.19 \times 10^{-52}$), and year effects are jointly significant once industry effects are included ($p = 0.026$), which supports the two-way specification. Year effects are not significant without industry effects, which suggests that year-to-year variation becomes detectable only after persistent differences across industries are removed.

*Hausman test.* A Hausman test does not reject the null that the random-effects estimator is consistent ($\chi^2 = 9.315$, $p = 0.157$). I use fixed effects nonetheless, because AI exposure is likely correlated with persistent industry characteristics such as capital intensity, skill composition, and

sectoral structure, and because the conventional Hausman test assumes that random effects is efficient, which does not hold when errors are serially correlated.

*Serial correlation.* A Breusch–Godfrey test strongly rejects the null of no serial correlation ($p = 9.20 \times 10^{-35}$), as expected for a persistent series. All standard errors are therefore clustered by industry (Arellano 1987), which makes inference robust to serial correlation within industries.

*Cross-sectional dependence.* The Pesaran (2021) CD test does not reject cross-sectional independence of the residuals ($CD = -0.737$, $p = 0.461$). After year fixed effects, there is therefore little evidence of productivity shocks spilling across industries, which industry-clustered standard errors would not account for.

*Multicollinearity.* The within-transformed correlations and variance inflation factors for the preferred specification show no meaningful collinearity among the regressors (Appendices I and II).

*Wald tests on the AI coefficients.* In column 5, the two AI coefficients are not jointly significant ($\chi^2 = 2.299$, $p = 0.317$). A test that the Early and Mature coefficients are equal also fails to reject, but because both are estimated imprecisely, the test has little power and does not show that the two stages have similar effects.

**5. Conclusion**

This paper asks whether U.S. industries more exposed to AI saw faster labor productivity growth as AI capabilities improved. I combine a time-varying, industry-level measure of AI exposure with an economy-wide index of AI capability in a two-way fixed-effects panel of 59 industries from 1997 to 2024. In the preferred specification, the coefficients on both early-stage and

mature-stage exposure are statistically indistinguishable from zero, and their confidence intervals are wide enough to include both sizable gains and sizable losses. Significant positive estimates appear only in first differences without year effects and in dynamic specifications with a lagged dependent variable, which imply implausibly large long-run effects; I do not treat them as evidence of an effect.

The absence of a detectable effect fits early economy-wide evidence, which finds no measurable effects of generative AI on earnings or hours worked (Humlum and Vestergaard 2025), even though experiments and field studies find large productivity gains on individual tasks (Noy and Zhang 2023; Brynjolfsson, Li, and Raymond 2025). It is also consistent with the argument that transformative technologies raise measured productivity only after costly complementary investments (Brynjolfsson, Rock, and Syverson 2019).

The analysis has four main limitations. First, the ability scores are fixed at their 2020 O*NET values, although the abilities most relevant to AI may change over time. Second, the employment shares used to build industry exposure are updated each year, so exposure is not strictly predetermined, and I do not test for pre-trends. Third, the mature-stage coefficient is identified mainly from 2022–2024, a short window that follows the pandemic, and the remote-work control covers only 2020–2021. Fourth, the results apply to the United States. Future work could hold employment shares at their 2016 values, estimate year-by-year effects to test for pre-trends, and use dynamic-panel methods. As data for later years become available, the effect of generative AI on industry productivity can be estimated more precisely.

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## 7. Data and Code Repository

**The Effect of AI Exposure on Labor Productivity**

## 8. Appendix

### I. Pairwise Correlation Matrix, Preferred Specification

| | Early × Agg × Intel | Mature × LM × Intel | ICT-capital contrib. | Non-ICT capital contrib. | Energy contrib. | WFH index × COVID dummy |
|---|---|---|---|---|---|---|
| Early × Agg × Intel | 1.000 | -0.124 | 0.129 | -0.026 | 0.005 | -0.089 |
| Mature × LM × Intel | -0.124 | 1.000 | 0.201 | -0.084 | 0.068 | 0.177 |
| ICT-capital contrib. | 0.129 | 0.201 | 1.000 | 0.122 | 0.077 | 0.133 |
| Non-ICT capital contrib. | -0.026 | -0.084 | 0.122 | 1.000 | -0.106 | -0.083 |
| Energy contrib. | 0.005 | 0.068 | 0.077 | -0.106 | 1.000 | 0.032 |
| WFH index × COVID dummy | -0.089 | 0.177 | 0.133 | -0.083 | 0.032 | 1.000 |

Within-transformed pairwise correlations between the regressors of the preferred static specification (Table 1, column 9). All correlations are modest ($|r| \leq 0.21$), indicating no meaningful collinearity among the regressors.

## II. Variance Inflation Factors, Preferred Specification

Variance-inflation factors (LSDV equivalent)

| Variable | GVIF | Df | GVIF^(1/(2*Df)) | Verdict |
|---|---|---|---|---|
| Early × Agg × Intel | 1.185 | 1 | 1.088 | Fine |
| Mature × LM × Intel | 1.270 | 1 | 1.127 | Fine |
| ICT-capital contrib. | 2.205 | 1 | 1.485 | Fine |
| Non-ICT capital contrib. | 1.943 | 1 | 1.394 | Fine |
| Energy contrib. | 2.736 | 1 | 1.654 | Fine |
| WFH index × COVID dummy | 3.088 | 1 | 1.757 | Fine |

Generalized variance inflation factors (LSDV equivalent) for the regressors in the preferred specification. All values are well below the conventional threshold of 5.

## III. Pairwise Correlation Matrix, Dynamic Specification

Within-transformed pairwise correlations of regressors in column (7)

| | early_x_agg_x_intel_raw | mature_x_lm_x_intel_raw | ict_capital_contrib | non_ict_capital_contrib | eng_contrib | lag(ln_lp, 1) | log_employment_lag1 | wfh_x_covid |
|---|---|---|---|---|---|---|---|---|
| early_x_agg_x_intel_raw | 1.000 | -0.537 | -0.217 | 0.091 | -0.106 | -0.105 | -0.047 | -0.360 |
| mature_x_lm_x_intel_raw | -0.537 | 1.000 | 0.306 | -0.071 | 0.167 | 0.158 | 0.075 | -0.026 |
| ict_capital_contrib | -0.217 | 0.306 | 1.000 | 0.213 | 0.132 | 0.277 | -0.036 | 0.013 |
| non_ict_capital_contrib | 0.091 | -0.071 | 0.213 | 1.000 | 0.142 | 0.193 | -0.158 | -0.143 |
| eng_contrib | -0.106 | 0.167 | 0.132 | 0.142 | 1.000 | 0.323 | -0.099 | 0.059 |
| lag(ln_lp, 1) | -0.105 | 0.158 | 0.277 | 0.193 | 0.323 | 1.000 | -0.262 | 0.009 |
| log_employment_lag1 | -0.047 | 0.075 | -0.036 | -0.158 | -0.099 | -0.262 | 1.000 | 0.023 |
| wfh_x_covid | -0.360 | -0.026 | 0.013 | -0.143 | 0.059 | 0.009 | 0.023 | 1.000 |

Within-transformed pairwise correlations for the dynamic specification in Table 1, column 7, which adds lagged log labor productivity and lagged log employment. Correlations are higher than in the preferred specification; the largest is between the two AI terms (−0.537).

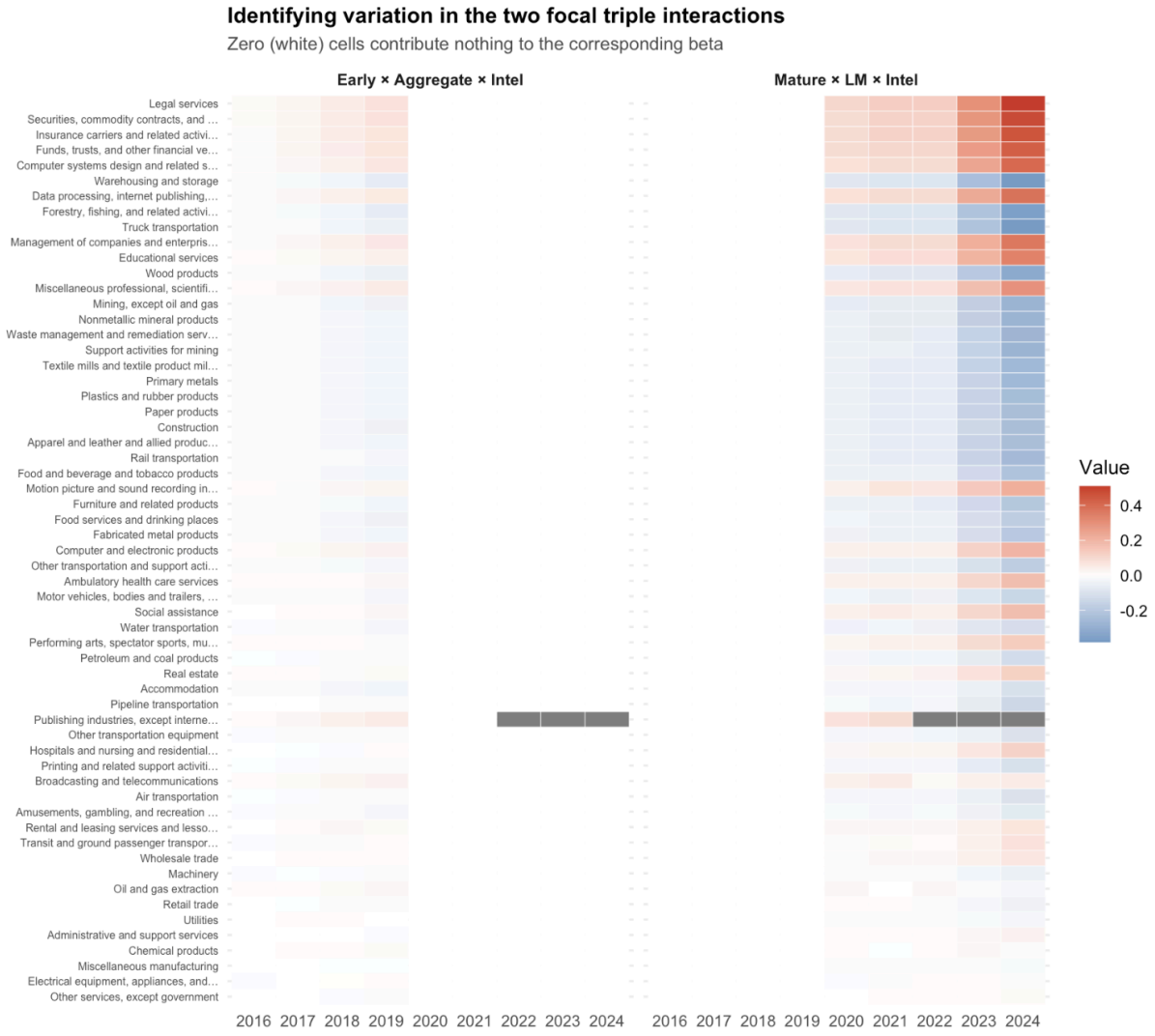
## IV. Variance Inflation Factors, Dynamic Specification

Variance-inflation factors (LSDV equivalent)

| Variable | GVIF | Df | GVIF^(1/(2·Df)) | Verdict |
|---|---|---|---|---|
| Early × Agg × Intel | 2.647 | 1 | 1.627 | Fine |
| Mature × LM × Intel | 2.745 | 1 | 1.657 | Fine |
| ICT-capital contrib. | 3.013 | 1 | 1.736 | Fine |
| Non-ICT capital contrib. | 2.005 | 1 | 1.416 | Fine |
| Energy contrib. | 3.629 | 1 | 1.905 | Fine |
| ln(LP)_{t-1} | 2.049 | 1 | 1.432 | Fine |
| Log employment_{t-1} | 56.441 | 1 | 7.513 | Problem |
| WFH index × COVID dummy | 4.021 | 1 | 2.005 | Fine |

Generalized variance inflation factors for the dynamic specification in Table 1, column 7. Lagged log employment has a GVIF of 56.4, indicating severe collinearity. Together with the bias from the lagged dependent variable (Nickell 1981), this motivates a dynamic-panel GMM estimator for any dynamic extension.

## V. Heatmap of Identifying Variation in the Two AI Interaction Terms




The heatmap shows which industry-year cells contribute variation to the Early × Aggregate AIIE × Intel and Mature × LM AIIE × Intel interactions; white cells contribute nothing. For the Mature term, most of the variation comes from 2022–2024, so its coefficient is identified from a short window and should become more precise as data for later years become available. Gray cells are missing data (publishing industries, 2022–2024).

## VI. Variance Inflation Factors, Two-Way FE Model with Both Lags

Variance-inflation factors for the two-way FE model with both lags

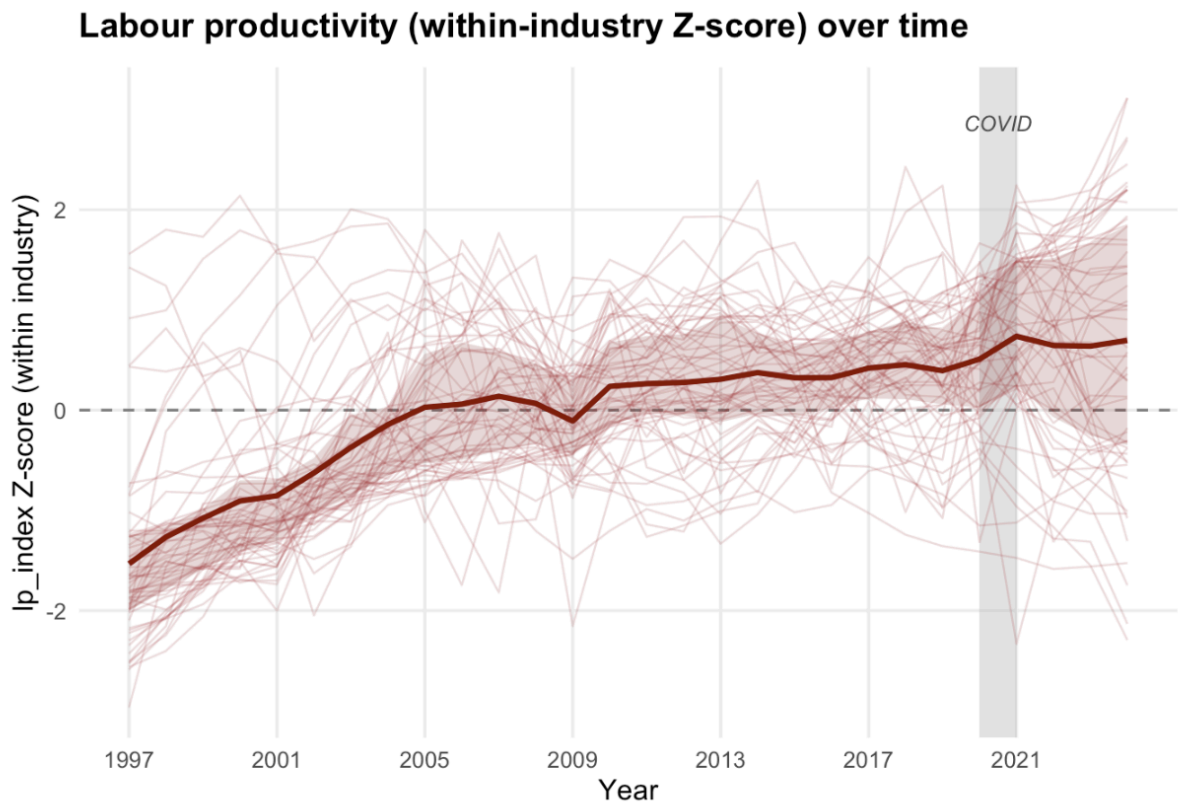
| Variable | GVIF | Df | GVIF^(1/(2·Df)) | Verdict |
|---|---|---|---|---|
| Early × Aggregate AIIE × Intel | 2.142 | 1 | 1.464 | No problem |
| Mature × LM AIIE × Intel | 2.554 | 1 | 1.598 | No problem |
| ICT-capital contribution | 3.072 | 1 | 1.753 | No problem |
| Non-ICT capital contribution | 1.958 | 1 | 1.399 | No problem |
| Energy contribution | 3.902 | 1 | 1.975 | No problem |
| ln(LP) t-1 | 3.352 | 1 | 1.831 | No problem |
| Log employment t-1 | 55.824 | 1 | 7.472 | Problem - do not include with final model |
| Industry fixed effects | 4009.306 | 58 | 1.074 | No problem |
| Year fixed effects | 1.835 | 8 | 1.039 | No problem |

*Notes:*

Threshold rules of thumb (on the GVIF^(1/(2·Df)) scale): < 2.24 = fine, 2.24–3.16 = noteworthy, > 3.16 = redundant.
Industry FE has a large raw GVIF by construction (59 levels); the scaled column is what's comparable across rows.
Log employment t-1 is flagged as redundant with the industry fixed effects, not with the other regressors.

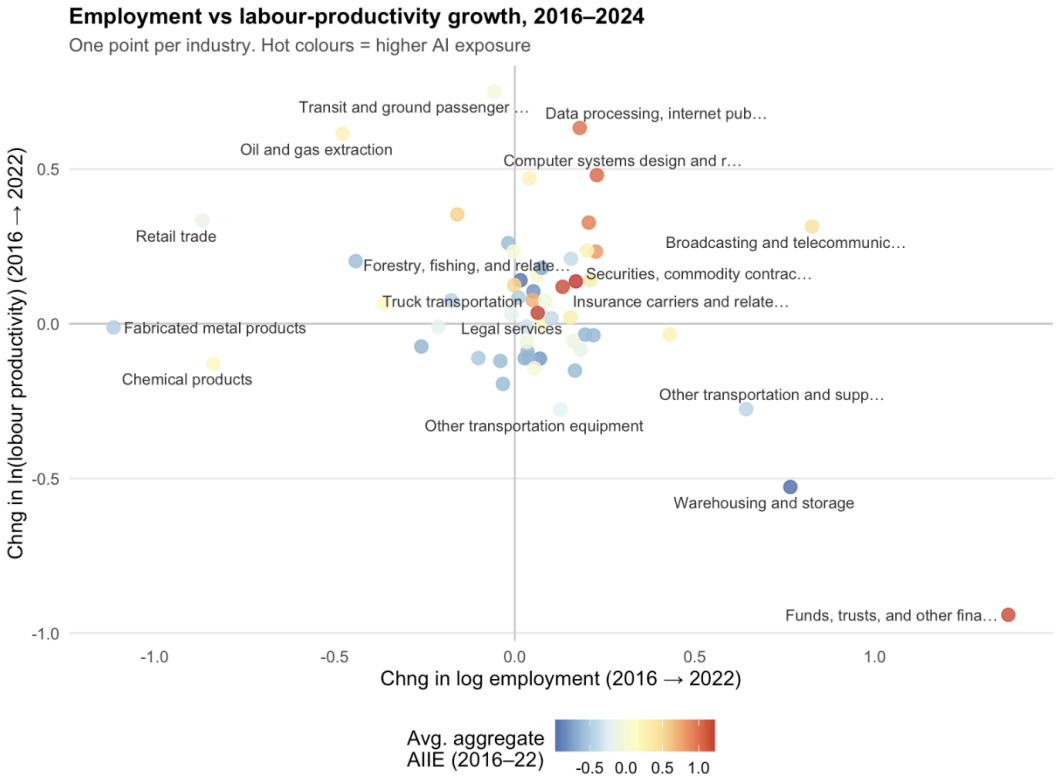
Generalized variance inflation factors for the two-way fixed-effects model with both lagged log labor productivity and lagged log employment. Lagged employment (GVIF = 55.8) is nearly collinear with the industry fixed effects, so it cannot be retained alongside them in a standard fixed-effects model.

**VII. Labor Productivity (Within-Industry Z-Score), 1997–2024**



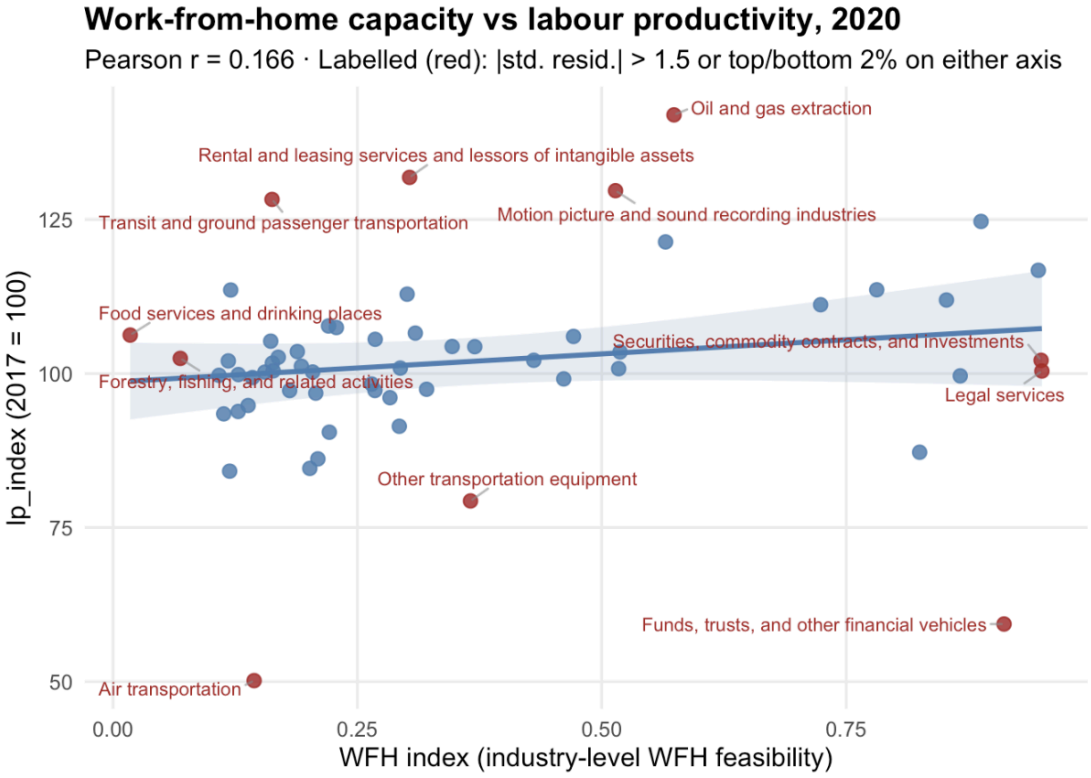

Each industry's labor productivity is standardized using its own mean and standard deviation over 1997–2024, so zero represents the industry's long-run average. The common upward trend, strongest from 1997 to the mid-2000s, motivates the year fixed effects, and the greater dispersion across industries after 2020 motivates the WFH × COVID control.

## VIII. Employment vs. Labor Productivity Growth by Industry, 2016–2022




Each point is one industry, colored by its average Aggregate AIIE over 2016–2022, with warmer colors indicating higher exposure; both axes show changes from 2016 to 2022. Some highly exposed industries, such as data processing and computer systems design, had strong productivity growth, while others, such as funds, trusts, and other financial vehicles, saw large declines. The figure is descriptive and does not control for other factors.

## IX. Work-From-Home Feasibility vs. Labor Productivity, 2020




Each point is one industry. Work-from-home feasibility and labor productivity in 2020 are only weakly correlated across industries (Pearson r = 0.166), and the correlation is not statistically significant for a sample of this size. Industries with unusual values, such as oil and gas extraction and air transportation, are labeled. The figure motivates the WFH control but does not by itself show that remote work raised productivity.